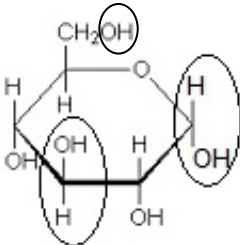


4(a)	converted the measured length (in mm) to μm (and dropped the, μm / units) <i>or</i> multiplied the measurement by 1000 ;	1
4(b)	phosphate(-containing) heads are hydrophilic <i>or</i> triglycerides do not have a hydrophilic portion ; lipid droplet is stored in / phosphate heads can interact with, cytosol / aqueous environment / watery environment ;	2
4(c)	to, digest / break down, worn out organelles / waste metabolic products / toxic substances / AW ; A invading pathogens	1

Question	Answer	Marks
4(d)(i)	correct orientation of H and OH on, C1 <u>and</u> on C3 ; OH added to carbon 6 ; 	2
4(d)(ii)	positive result is coloured precipitate ; <i>only allow if hydrolysis noted</i> A green / yellow / orange / brown / red, <i>for colour</i> <i>any two from</i> <i>ref. to</i> (heat with) Benedict's (reagent / solution) and, negative test / no colour change / remains blue, and test again with Benedict's ; boil with (hydrochloric) acid ; A hydrolyse with, acid / enzyme AVP ; e.g. use a fresh sample to hydrolyse boil for 5-10 minutes with acid <u>cool</u> before neutralising neutralise (remaining acid) with alkali test with universal indicator paper (to check for pH7) compare with a control <i>allow one mark for heat with Benedict's and coloured precipitate</i>	3

Question	Answer	Marks
4(e)(i)	any two from (higher K_m enzyme) has a lower affinity for its substrate ; A binds substrate less easily needs a higher concentration of substrate to reach, V_{max} / maximum activity / $\frac{1}{2} V_{max}$; less likely to be saturated with substrate ; variations in substrate have greater effect on rate of reaction ;	2
4(e)(ii)	any two from in the vacuole ; A in lysosome qualified ; e.g. has, acidic / low pH, environment contains, acid hydrolases / enzymes that require low pH non-regulatory trehalase needs acidic conditions for optimum activity cytosol has neutral pH so likely to be location of regulatory trehalase (and enzymes are in different locations) if in cytosol then this would mean low pH and other enzymes, would (partially) denature / work below optimum as acid conditions required, will be in area protected from rest of cell and vacuole has the tonoplast as barrier A lysosomes are membrane bound	2
4(e)(iii)	yes because they both work, within the cell / inside the cell ;	1

Question	Answer	Marks
4(f)	<i>any three from</i> <i>likely to be regulatory trehalase / unlikely to be non-regulatory trehalase</i> nearer to optimum of regulatory trehalase (of <i>S. cerevisiae</i>) ; ora pH 6.5 / 6.6 nearer to pH 7.0 (than to pH 5.0 of non-regulatory trehalase) ; no / (very) low, activity at pH 4.5 <i>or</i> greater activity at pH 7 than pH 4.5(–5.0) ; <i>alternative</i> suggestion that enzyme could be a different form of trehalase ; because has different optimum pH to both regulatory and non-regulatory ; pH 6.5 / 6.6, rather than pH 7.0 or pH 5.0 ; <i>general</i> (so) likely to act in the cytosol with, neutral pH / pH7 ; unlikely to be found in vacuole / lysosome, with low / acid, pH ;	3